

Counterfactual Oracle Values for PPO

Literature Review, Pitch, and Experimental Plan

Abstract

This document proposes a deliberately narrow study of two questions. First, does access to exact, policy-relative counterfactual action values improve PPO learning when ordinary advantage estimation suffers from delayed and noisy credit assignment? Second, can the same information introduce a secondary objective into a competent policy while preserving its original goal? The study uses only exact simulator/dynamic-programming oracle values; it does not implement COCOA, HCA, or any learned counterfactual estimator. The oracle is privileged information and a feasibility upper bound, not a deployable method.

1 Research questions and intended claims

Let $J_g(\pi)$ denote a primary goal, such as collecting a treasure or finishing a level. Let $J_c(\pi)$ denote a secondary behaviour, such as collecting optional items, and let $J_e(\pi)$ denote a step or energy cost.

H1: oracle counterfactual credit. Exact all-action counterfactual values improve PPO learning on delayed, high-variance credit-assignment tasks. This is an information-value claim: the oracle supplies better action credit than ordinary trajectory-based PPO.

H2: goal-preserving preference adaptation. Starting from a policy already competent on J_g , exact counterfactual goal values identify local action slack. Steering only within this slack can increase J_c while limiting the loss in J_g , and can outperform naive reward mixing at a matched primary-goal constraint.

The study does *not* claim a new general counterfactual estimator, a universal PPO improvement, or a guaranteed safety constraint. Local gating is an empirical mechanism; global preservation must be measured.

2 Literature review and research boundary

The policy-gradient theorem gives

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{s \sim d^{\pi}, a \sim \pi_{\theta}} [A^{\pi}(s, a) \nabla_{\theta} \log \pi_{\theta}(a \mid s)]. \quad (1)$$

PPO uses a clipped surrogate objective (Schulman et al., 2017); in practice, its actor weights are usually Generalized Advantage Estimates (GAE) (Schulman et al., 2016). GAE trades bias and variance, but delayed rewards, future-policy randomness, and unrelated stochastic rewards can make sampled action credit noisier than the conditional action value.

State-only baselines have zero population score-gradient contribution. Thus, action-selective credit is standard policy-gradient algebra, not the novelty of this study. Action-dependent factorized baselines already provide unbiased variance reduction in structured policies (Wu et al., 2018).

Long-horizon credit assignment is well studied. RUDDER redistributes delayed returns (Arjona-Medina et al., 2019). Hindsight Credit Assignment (HCA) and Counterfactual Credit Assignment (CCA) use future information to distinguish skill from luck (Harutyunyan et al., 2019; Mesnard et al., 2021). COCOA estimates contributions to rewarding outcomes rather than merely to future states (Meulemans et al., 2023). This project stops before the learned-estimator problem: it asks whether perfect counterfactual values are useful enough to justify later approximation work.

The secondary-objective problem is related to constrained and multi-objective RL:

$$\max_{\pi} J_c(\pi) \quad \text{subject to} \quad J_g(\pi) \geq J_g(\pi_0) - \epsilon. \quad (2)$$

Constrained RLHF has used Lagrange multipliers to constrain component reward models (Moskovitz et al., 2023). The desired contribution here is more local: whether counterfactual action values can restrict secondary steering to actions with low primary-goal opportunity cost.

3 Exact counterfactual oracle

For a fixed rollout policy π_{old} and fully observed state s , define

$$Q_g^{\pi_{\text{old}}}(s, a) = \mathbb{E} \left[\sum_{\ell \geq 0} \gamma^\ell r_{t+\ell}^g \mid s_t = s, \text{ do}(a_t = a), \pi_{\text{old}} \text{ thereafter} \right]. \quad (3)$$

In the linear environments, this quantity is computed exactly by dynamic programming. In the MiniGrid transfer it is estimated by branching from saved simulator states and averaging fixed-continuation rollouts; that quantity is called a *simulator oracle estimate*, not an exact value.

The centered counterfactual advantage is

$$A_{\text{CF},g}(s, a) = Q_g^{\pi_{\text{old}}}(s, a) - \sum_{a'} \pi_{\text{old}}(a' \mid s) Q_g^{\pi_{\text{old}}}(s, a'). \quad (4)$$

For a discrete policy, use an all-action PPO surrogate

$$L_{\text{CF}}(\theta) = -\frac{1}{|B|} \sum_{s \in B} \sum_a \pi_{\text{old}}(a \mid s) \min(r_\theta(s, a) A_{\text{CF},g}(s, a), \text{clip}(r_\theta(s, a), 1 - \epsilon, 1 + \epsilon) A_{\text{CF},g}(s, a)), \quad (5)$$

where $r_\theta(s, a) = \pi_\theta(a \mid s) / \pi_{\text{old}}(a \mid s)$. The oracle arm uses

$$L_{\text{actor}} = (1 - \alpha) L_{\text{PPO}} + \alpha L_{\text{CF}}, \quad \alpha = 0.5. \quad (6)$$

The critic is always trained on the original primary-goal return; oracle values never become critic targets.

4 Goal-preserving secondary steering

At adaptation time, let $Q_c(s, a)$ and $Q_e(s, a)$ be the oracle values for optional collection and cost. Define an action set with low primary-goal opportunity cost:

$$\mathcal{A}_\delta(s) = \left\{ a : Q_g(s, a) \geq \max_{a'} Q_g(s, a') - \delta \right\}. \quad (7)$$

The detached secondary direction is

$$d_\beta(s, a) = \mathbf{1}\{a \in \mathcal{A}_\delta(s)\} [\beta \Delta_c(s, a) - \eta \Delta_e(s, a)], \quad (8)$$

where $\Delta_i(s, a) = Q_i(s, a) - \sum_{a'} \pi_{\text{old}}(a' | s) Q_i(s, a')$. The gated arm optimizes an all-action PPO surrogate using d_β as its actor weight, together with a fixed KL penalty to π_0 . Equation (7) is a local proposal filter, not a proof of the global constraint in Equation (2).

5 Environments

Environment	Role	Why it is included
Linear Key-to-Door (L)	H1 oracle credit	Public credit-assignment benchmark. An early key action enables a later treasure reward; stochastic apple rewards are distractors; key-door distance L controls delayed-credit difficulty.
Key-to-Door-Collect (L)	H2 exact adaptation	Fixed modification: treasure is the sole base reward; apples are separately logged optional collectibles that cost time. Dynamic programming remains exact.
MiniGrid Collect (8×8)	DoorKey- H2 transfer	Fixed DoorKey layout with two optional collection tiles on detours. It tests grid navigation and a larger discrete action set while retaining the key-door-goal structure.

Table 1: Small environments used in the study. Key-to-Door-Collect and DoorKey-Collect will be released as fixed, documented variants.

6 Pilot experiments and gates

6.1 P1: Is oracle action credit useful?

Run Linear Key-to-Door with $L = 40$, a two-layer 64-unit MLP PPO policy, and five paired seeds. Each arm has a cap of 150,000 environment transitions.

1. Standard PPO–GAE.
2. PPO plus the exact all-action oracle objective in Equation (6).
3. PPO plus a state-shuffled oracle, where the same oracle vectors are assigned to different batch states.

The primary outcome is area under the treasure-success learning curve. Report final treasure success, apple reward, gradient squared error to the exact policy gradient, and gradient cosine with the exact policy gradient.

P1 gate. Advance only if the paired confidence interval for oracle-minus-PPO treasure-success AUC is positive and the shuffled arm does not reproduce the benefit. If the exact oracle does not help, do not pursue secondary steering as an algorithmic claim.

6.2 P2: Can an oracle add a secondary preference safely?

First train a goal-only base policy π_0 on Key-to-Door-Collect with $L = 40$. The base policy must reach stable treasure success before adaptation begins. Fork the identical checkpoint into:

1. no adaptation;
2. naive reward mixing $r_g + \beta r_c - \eta r_e$;
3. global constrained PPO for Equation (2);
4. counterfactual-gated steering from Equations (7)–(8);
5. state-shuffled counterfactual-gated steering.

Use $\beta \in \{0.5, 1.0\}$, five paired seeds, and 75,000 adaptation transitions per arm. The primary outcome is collectibles per successful episode subject to a maximum absolute treasure-success decrease of 0.05.

P2 gate. Advance only if a gated arm increases collection while respecting the goal-success budget, and reward mixing either causes a larger goal drop or requires more adaptation to achieve the same collection level.

7 Full modest-compute study

All full experiments use paired seeds, frozen pilot-selected hyperparameters, and seed-level confidence intervals.

7.1 E1: exact oracle credit across delay

Run Linear Key-to-Door at $L \in \{20, 80\}$ with the three P1 arms. Use eight paired seeds and 300,000 transitions per arm and difficulty:

$$2 \text{ difficulties} \times 3 \text{ arms} \times 8 \text{ seeds} \times 300,000 = 14.4 \text{ million transitions.}$$

The pre-registered claim is that oracle benefit is clearer in the longer-delay condition and that exact gradient alignment mediates the learning benefit. The oracle is reported as privileged information, not as a practical method.

7.2 E2: goal-preserving secondary adaptation

On Key-to-Door-Collect, train eight goal-only base policies and fork each into the five P2 arms. Use $\beta \in \{0.5, 1.0\}$ and 150,000 adaptation transitions per arm. The primary analysis is the Pareto frontier

$$\text{collectibles per successful episode} \quad \text{versus} \quad \text{primary treasure-success drop.}$$

The counterfactual method is successful only if it lies above reward mixing at the same goal-success loss and is competitive with constrained PPO. Increasing collection while destroying treasure success is a failed constraint result.

7.3 E3: one gridworld transfer

On MiniGrid DoorKey-Collect, run only reward mixing, constrained PPO, and counterfactual-gated steering, with eight paired seeds and 200,000 adaptation transitions per arm. Use a fixed number of simulator branches from saved states to estimate goal action values. This transfer is intentionally small: it tests whether the preference result survives beyond the linear environment without becoming a benchmark sweep.

The full study is approximately 30–35 million small, discrete-environment transitions, plus pilots. It is CPU-scale; simulator branching in E3 is the only potentially expensive operation.

8 Implementation and reporting rules

1. Recompute oracle values for the current rollout policy; Q^π is policy-dependent.
2. Detach oracle values from actor gradients and freeze them throughout PPO minibatch epochs.
3. Keep critic targets tied to the original primary environment return.
4. Use identical initialization, environment seeds, rollout lengths, and advantage-scale conventions across paired arms.
5. Record ordinary environment transitions separately from simulator branches. The oracle is an upper bound, not a free interaction budget.
6. Select α , β , η , and δ in pilots only; freeze them before E1–E3.
7. Report goal success, optional collection, and cost separately rather than only a scalar return.

9 Pitch

The project tests whether exact counterfactual action values expose useful *slack* in a learned policy. In the first setting, the oracle reveals which actions genuinely contribute to a delayed primary goal despite noisy distractor returns. In the second, it identifies actions that can pursue a newly introduced secondary preference without paying a large primary-goal opportunity cost. The intended final statement is:

Exact counterfactual action values can improve PPO’s credit signal in delayed, high-variance tasks and can support controlled post-training preference adaptation. Their value comes not from arbitrary advantage perturbation, but from restricting secondary policy changes to actions whose counterfactual primary value remains high.

References

- J. A. Arjona-Medina, M. Gillhofer, M. Widrich, T. Unterthiner, J. Brandstetter, and S. Hochreiter. RUDDER: Return decomposition for delayed rewards. In *NeurIPS*, 2019.
- A. Harutyunyan, W. Dabney, T. Mesnard, M. Azar, P. Piot, N. Heess, H. van Hasselt, G. Wayne, S. Singh, and M. Precup. Hindsight credit assignment. In *NeurIPS*, 2019.
- A. Meulemans, S. Schug, S. Kobayashi, N. Daw, and G. Wayne. Would I have gotten that reward? Long-term credit assignment by counterfactual contribution analysis. In *NeurIPS*, 2023.

- T. Mesnard, T. Weber, F. Viola, S. Thakoor, A. Saade, A. Harutyunyan, W. Dabney, N. Heess, A. Guez, E. Moulines, M. Hutter, L. Buesing, and R. Munos. Counterfactual credit assignment in model-free reinforcement learning. In *ICML*, 2021.
- T. Moskovitz, A. K. Singh, D. J. Strouse, T. Sandholm, R. Salakhutdinov, A. D. Dragan, and S. McAleer. Confronting reward model overoptimization with constrained RLHF. arXiv:2310.04373, 2023.
- J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel. High-dimensional continuous control using generalized advantage estimation. In *ICLR*, 2016.
- J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. arXiv:1707.06347, 2017.
- C. Wu, A. Rajeswaran, Y. Duan, V. Kumar, A. Bayen, S. Kakade, I. Mordatch, and P. Abbeel. Variance reduction for policy gradient with action-dependent factorized baselines. In *ICLR*, 2018.